RAG Learning Sample Document

1. What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with a language model. Before answering a question, the system searches a collection of documents and provides relevant passages to the model as context.

2. Why do we use document chunking?

Chunking divides a long document into smaller pieces of text. Smaller chunks are easier to embed, search, and pass into a model's context window. A chunk should be large enough to preserve meaning but small enough to retrieve focused information.

3. What are embeddings?

An embedding is a list of numbers that represents the meaning or features of text. Texts with similar meanings often have vectors that are close together in embedding space. A vector database can store these vectors and retrieve semantically similar passages.

4. Example Python learning task

Load this PDF in Python, extract its text, clean the text, and split it into chunks. Then inspect each chunk before moving on to embeddings.